# Hypotheses

*What do we currently believe explains the problem, and how could we prove ourselves wrong?*



# Today

By the end of today, you will have

**Written a hypothesis register with disconfirming evidence and rival explanations, and defended it in the Hypothesis Battle.**

- Write three to five testable hypotheses from your priority branches.
- State, for each, the evidence that would weaken it.
- Name a rival explanation and the evidence that would tell the two apart.
- Separate assumptions from hypotheses and decide which to test first.

**Competencies: F1 · F2 · F3**

Session plan

| | |
|---|---|
| 10 min | **Opening problem** |
| 15 min | **Concept: testable hypotheses** |
| 15 min | **Demonstration: Madhyapur register** |
| 45 min | **Thursday challenge: hypothesis battle** |
| 25 min | **Own case: hypothesis register** |
| 10 min | **Debrief and exit task** |

# Opening problem

- **Women's participation in paid work is low in a district. Write the explanation you find most convincing, in one sentence.**

Now write the evidence that would prove it wrong. Which sentence was harder to write?

# A hypothesis is a claim evidence could show to be wrong

## Not testable

- Social norms are a factor
- Awareness needs improvement
- Many issues contribute

## Testable

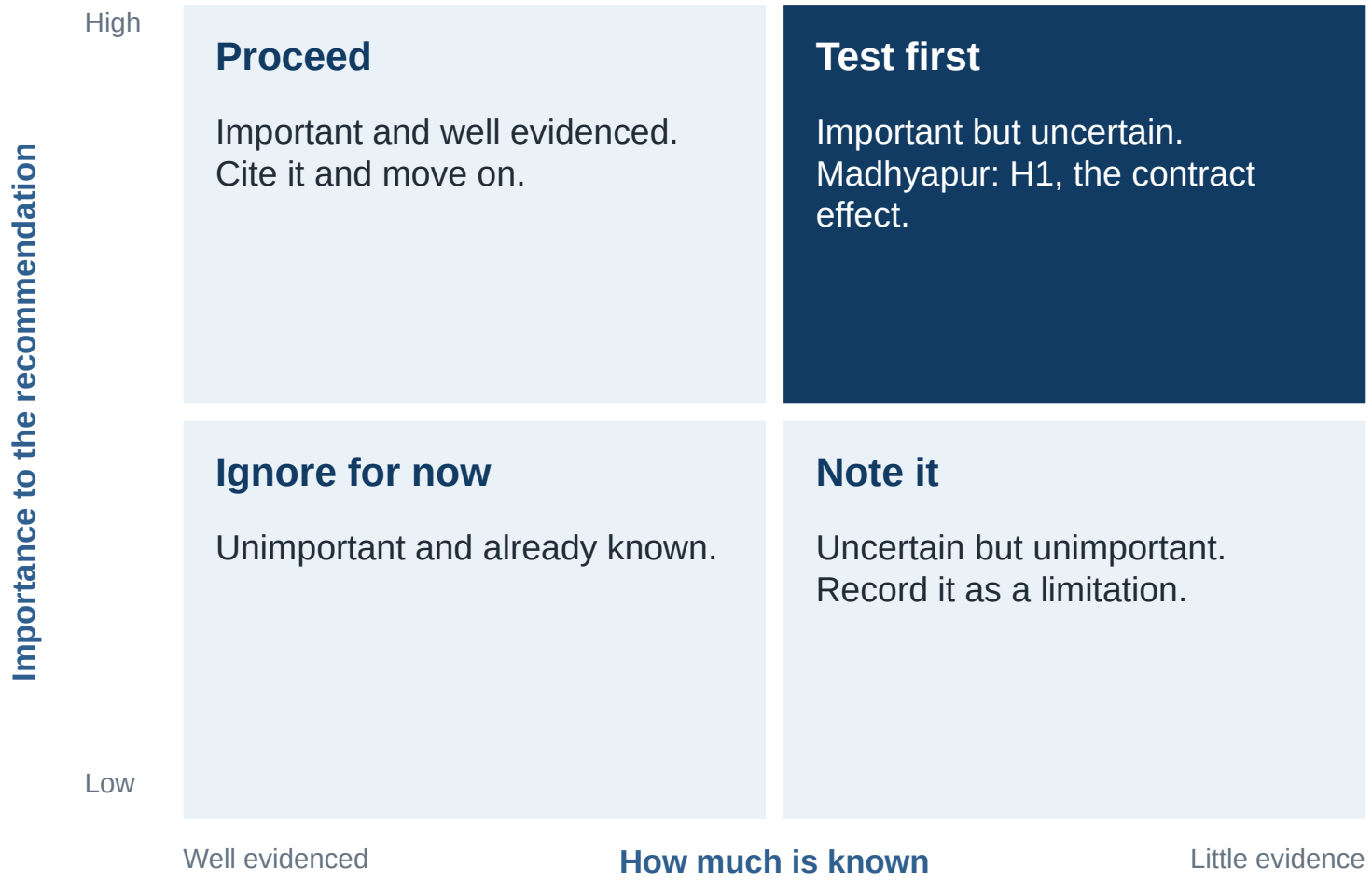
- Where safe transport is available, more women in the same district take paid work
- Households that separate waste report crews keeping streams apart
- Wards with two-compartment vehicles show more separation than similar wards without

**If no evidence could count against it, it is not a hypothesis.**

# The Madhyapur hypothesis register

| # | Hypothesis | Evidence that would weaken it | Rival explanation |
|---|---|---|---|
| **H1** | Payment by weight leads crews to mix separated waste | Crews mix just as much under per-household or per-trip contracts | Vehicles lack compartments, whatever the payment |
| **H2** | Households lack a second container and space | Separation is no higher where bins were distributed | Households have containers but see no point |
| **H3** | With no dry-waste processing, separation is pointless downstream | Separated dry waste is in fact recovered by waste pickers | Processing exists but is not linked to collection |
| **H4** | Penalties are not applied, so the rules are not felt | Separation is no higher in wards where fines were issued | Fines raise dumping, not separation |
| **H5** | Households are unsure what counts as dry waste | Households sort correctly when asked | They know how but see no reason |

# Assumption mapping: which beliefs to test first




Assumption mapping is used widely in design and product practice to decide which beliefs to test before committing resources. Here it orders the workplan.

# Guarding against confirmation bias

1. **Write the disconfirming test first**

   Before looking for support, say what would count against the hypothesis.

2. **Keep a rival for every hypothesis**

   Ask what else would produce the same observation.

3. **Look where you expect to be wrong**

   Search for the comparison most likely to embarrass your favourite.

4. **Label assumptions separately**

   Mark beliefs you are not testing, so they stay visible.

# Activity: The hypothesis battle

Defend a hypothesis against rivals, and learn to specify evidence that would tell them apart.

### Scenario

Women's participation in paid work is low in a stylised district with growing towns. Each pod defends one explanation: care responsibilities; a lack of suitable local jobs; social norms about women's work; safety and transport; a mismatch of skills and education; or household income. Other pods will try to knock it down.

### What to do

1. Turn your explanation into one testable hypothesis.
2. Write the evidence that would support it and the evidence that would weaken it.
3. Present it in 90 seconds.
4. Each other pod gets one challenge: a rival explanation or a disconfirming test.
5. Revise, and name the evidence that would decide between you and your strongest rival.

**Time** 45 minutes (Thursday challenge)

**Groups** Six pods or pairs, one explanation each

**Output** Six sharpened hypotheses with disconfirming tests

# Your case: Worksheet 4.2

## Hypothesis register

1. Turn each priority branch into one or two testable hypotheses, three to five in all.
2. For each, write the evidence that would weaken it and where you might find it.
3. Name a rival explanation and the evidence that would distinguish the two.
4. Map your assumptions by importance and evidence; mark which to test first.
5. Rate your current confidence in each: low, medium or high.

**Hand in**

Worksheet 4.2 by the start of Day 10, for Gate 2.

**Quality check**

Three to five falsifiable hypotheses.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

# What to take away

1. A hypothesis is a claim that evidence could show to be wrong.
2. Write the disconfirming test before looking for support.
3. Keep a rival for every hypothesis; several may be partly true.
4. Test first what is important and uncertain.

**Exit task** Write the single piece of evidence that would most damage your favourite hypothesis.

# Before the next session

### Independent work

For each hypothesis, find one published source that could test it, and check that it covers your place and period. About 45 minutes.

### Read

Bardach and Patashnik, A Practical Guide for Policy Analysis, part I, step two: assemble some evidence.

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### Next

## Day 10: The workplan

Tomorrow you turn the register into a workplan, open your evidence register and submit Gate 2.